

Kubernetes Lab: Program 10 - Minikube

Deployment and Dashboard

This document details the configuration and execution steps for deploying an NGINX application using Minikube and monitoring it via the Kubernetes Dashboard.

Project Structure

```
Program-10/  
├── deployment.yaml  
└── service.yaml
```

Configuration Files

1. Deployment Configuration (deployment.yaml)

This file defines a deployment that runs two replicas of the NGINX web server.

```
apiVersion: apps/v1  
kind: Deployment  
metadata:  
  name: nginx-deployment  
spec:  
  replicas: 2  
  selector:  
    matchLabels:  
      app: nginx  
  template:  
    metadata:  
      labels:  
        app: nginx  
    spec:  
      containers:  
        - name: nginx  
          image: nginx:latest  
          ports:  
            - containerPort: 80
```

2. Service Configuration (service.yaml)

This file defines a NodePort service to expose the NGINX deployment to external traffic.

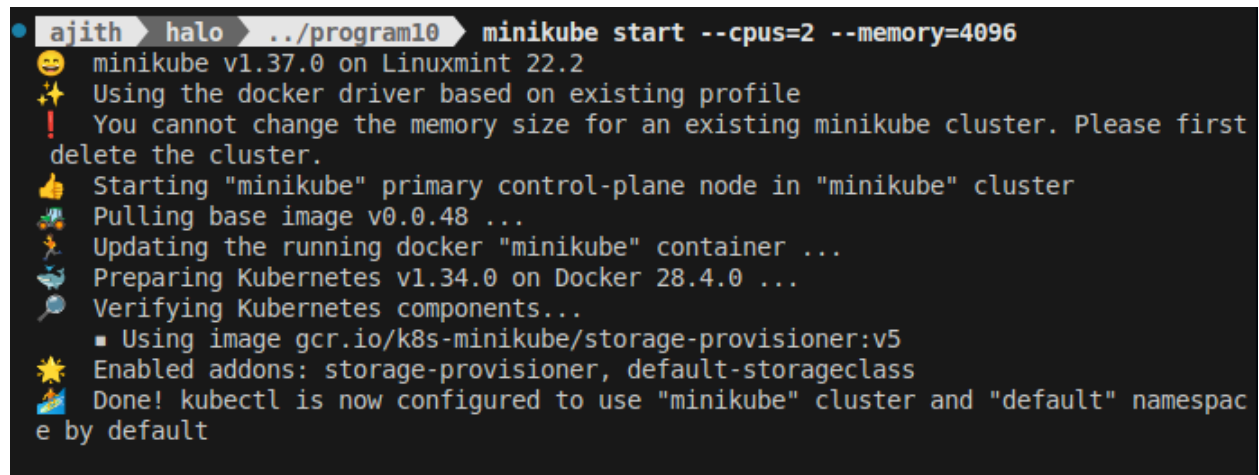
```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Execution Steps

Step 1: Initiate Minikube

Start the Minikube cluster with specified CPU and memory allocation.

```
minikube start --cpus=2 --memory=4096
```

A terminal window with a dark background. The prompt is 'ajith@halo: ~/program10'. The command entered is 'minikube start --cpus=2 --memory=4096'. The output shows minikube v1.37.0 on Linuxmint 22.2, using the docker driver. It warns that memory size cannot be changed for an existing cluster. It then starts the 'minikube' primary control-plane node, pulls base image v0.0.48, updates the docker container, prepares Kubernetes v1.34.0 on Docker 28.4.0, and verifies components. It uses the image gcr.io/k8s-minikube/storage-provisioner:v5 and enables addons: storage-provisioner, default-storageclass. Finally, it says 'Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default'.

Step 2: Enable Kubernetes Dashboard

Open the web-based UI to monitor your cluster.

```
minikube dashboard
```

```
ajith@halo:~/program10$ minikube dashboard
Enabling dashboard ...
  ■ Using image docker.io/kubernetesui/dashboard:v2.7.0
  ■ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features
please run:

    minikube addons enable metrics-server

😞 Verifying dashboard health ...
🚀 Launching proxy ...
😞 Verifying proxy health ...
🌐 Opening http://127.0.0.1:46221/api/v1/namespaces/kubernetes-dashboard/services/h
ttp:kubernetes-dashboard:/proxy/ in your default browser...
ATTENTION: default value of option mesa_glthread overridden by environment.
```

Step 3: Apply Resources

Create the deployment and the service.

```
kubectl apply -f deployment.yaml
```

```
kubectl apply -f service.yaml
```

```
ajith@halo:~/program10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment unchanged
```

```
ajith@halo:~/program10$ kubectl apply -f service.yaml
service/nginx-service created
```

Step 4: Monitor via CLI

Verify that the resources were created successfully.

```
# Check status of pods
```

```
kubectl get pods
```

```
# Check cluster nodes
```

```
kubectl get nodes
```

```
# Check services
```

```
kubectl get svc
```

```
ajith halo ../Devops_Lab/program9 kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
demo-deploy-6876f5fc77-8jdlf        1/1     Running   6 (6d22h ago)    17d
demo-pod1                            1/1     Running   5 (6d22h ago)    17d
hello-2s5jb                          0/1     Completed 0                17d
hello-775d79c56b-xfkxt              1/1     Running   5 (6d22h ago)    17d
hello2-8797688b4-q662v              1/1     Running   5 (6d22h ago)    17d
nginx-deployment-6f9664446b-kfddw    1/1     Running   1 (6d22h ago)    7d10h
nginx-deployment-6f9664446b-q586r    1/1     Running   1 (6d22h ago)    7d10h
program-8                            1/1     Running   1 (2m41s ago)    6d23h
yaml-demo                            1/1     Running   4 (6d22h ago)    17d

ajith halo ../program10 kubectl get nodes
NAME        STATUS   ROLES    AGE   VERSION
minikube    Ready   control-plane  18d   v1.34.0

ajith halo ../program10 kubectl get svc
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
demo-pod1           ClusterIP   10.111.123.37 <none>         80/TCP           17d
hello               NodePort    10.106.180.195 <none>         80:31133/TCP     17d
kubernetes           ClusterIP   10.96.0.1      <none>         443/TCP          18d
kubia               NodePort    10.99.113.129  <none>         8080:31489/TCP   10d
nginx-deployment     NodePort    10.96.62.106   <none>         80:31012/TCP     7d10h
nginx-service        NodePort    10.103.246.107 <none>         80:30559/TCP     61s
```

Step 5: Inspect Pods

Debug or view details of specific pods.

View pod logs

kubectl logs <pod-name>

Describe pod for detailed event information

kubectl describe pod <pod-name>

```
ajith halo ../program10 kubectl logs nginx-deployment-6f9664446b-q586r
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/19 04:56:34 [notice] 1#1: using the "epoll" event method
2025/12/19 04:56:34 [notice] 1#1: nginx/1.29.4
2025/12/19 04:56:34 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/19 04:56:34 [notice] 1#1: OS: Linux 6.14.0-33-generic
2025/12/19 04:56:34 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/19 04:56:34 [notice] 1#1: start worker processes
2025/12/19 04:56:34 [notice] 1#1: start worker process 29
2025/12/19 04:56:34 [notice] 1#1: start worker process 30
2025/12/19 04:56:34 [notice] 1#1: start worker process 31
2025/12/19 04:56:34 [notice] 1#1: start worker process 32
2025/12/19 04:56:34 [notice] 1#1: start worker process 33
2025/12/19 04:56:34 [notice] 1#1: start worker process 34
2025/12/19 04:56:34 [notice] 1#1: start worker process 35
2025/12/19 04:56:34 [notice] 1#1: start worker process 36
```

```

ajith@halo:~/program10$ kubectl describe pod nginx-deployment-6f9664446b-q586
Name: nginx-deployment-6f9664446b-q586r
Namespace: default
Priority: 0
Service Account: default
Node: minikube/192.168.49.2
Start Time: Thu, 11 Dec 2025 23:20:45 +0530
Labels: app=nginx
        pod-template-hash=6f9664446b
Annotations: <none>
Status: Running
IP: 10.244.0.164
IPs:
  IP: 10.244.0.164
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID: docker://d3116a89c9c6209f943f026a3b00540b4e966bfe5552ee6c98f34f74e2dfb934
    Image: nginx:latest
    Image ID: docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
    Port: 80/TCP
    Host Port: 0/TCP
    State: Running
      Started: Fri, 19 Dec 2025 10:26:34 +0530
    Last State: Terminated
      Reason: Completed
      Exit Code: 0
      Started: Fri, 19 Dec 2025 10:14:21 +0530
      Finished: Fri, 19 Dec 2025 10:24:12 +0530
    Ready: True
    Restart Count: 2
    Environment: <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-gbd9g (ro)
Conditions:
  Type                               Status
  PodReadyToStartContainers          True
  Initialized                         True
  Ready                              True
  ContainersReady                    True
  PodScheduled                       True
Volumes:
  kube-api-access-gbd9g:
    Type: Projected (a volume that contains injected data from multiple sources)

```

Step 6: Access the Application

Open the NGINX application in your browser using the Minikube service command.

```
minikube service nginx-service
```

```

ajith@halo:~/program10$ minikube service nginx-service

```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:30559

```

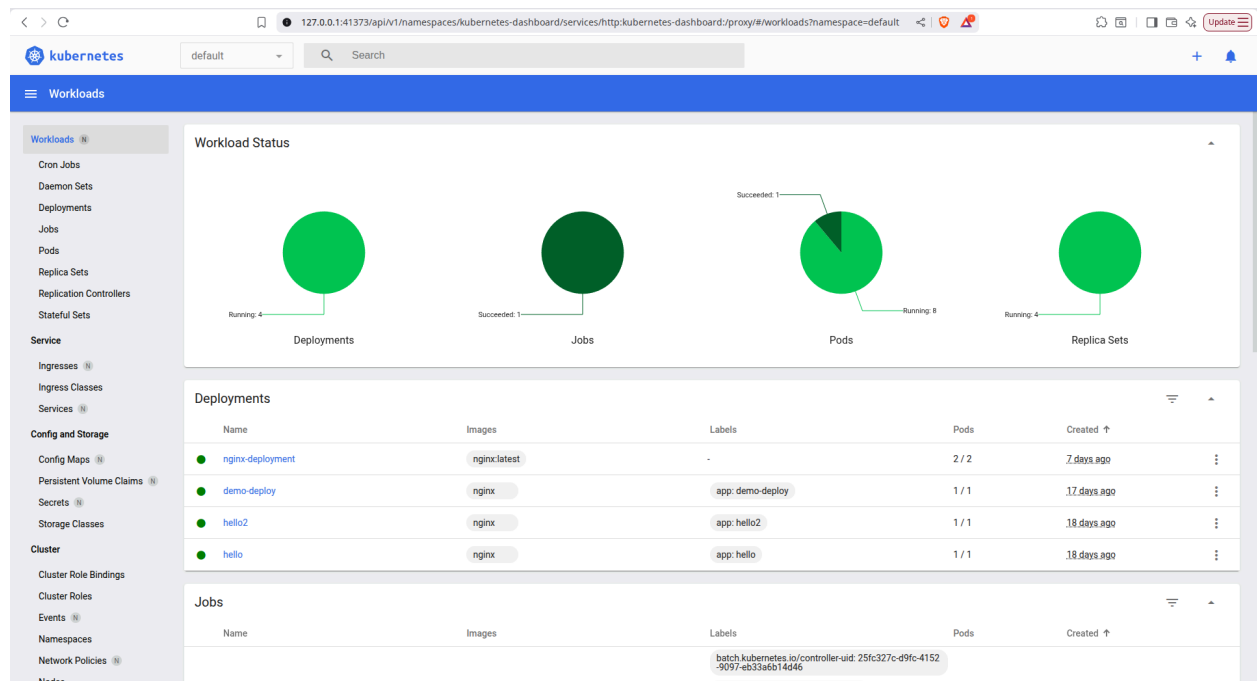
🚀 Opening service default/nginx-service in default browser...
ATTENTION: default value of option mesa_glthread overridden by environment.

```

```

ajith@halo:~/program10$ minikube dashboard
👉 Verifying dashboard health ...
👉 Launching proxy ...
👉 Verifying proxy health ...
👉 Opening http://127.0.0.1:41373/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
ATTENTION: default value of option mesa_glthread overridden by environment.

```



🔒 Not secure 192.168.49.2:30559

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Step 7: Cleanup

Delete the created resources to clean up the cluster.

```
# Delete all services  
kubectl delete svc --all
```

```
# Delete all deployments  
kubectl delete deployments --all
```